

Thesis Project

Building an Analytical Data Warehouse



**Completed by:
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Project Goals and Objectives

Slide 2

Goal: To build an analytical data warehouse for the store.



Objectives:

- ☐ Data generation and loading
- ☐ Creation of the data warehouse structure
- ☐ Development of ETL processes
- ☐ Visualization of key metrics

Project Architecture

NDS (Normalized Data Storage):
Layer for storing normalized data
according to 3NF

Stage: Layer for storing raw
data after loading from
source files (CSV)

DDS (Data Dimensional Storage): Analytics
layer, organized as a star schema with fact
and dimension tables

DQ (Data Quality): Layer
responsible for data quality
control, including checks and
cleansing.

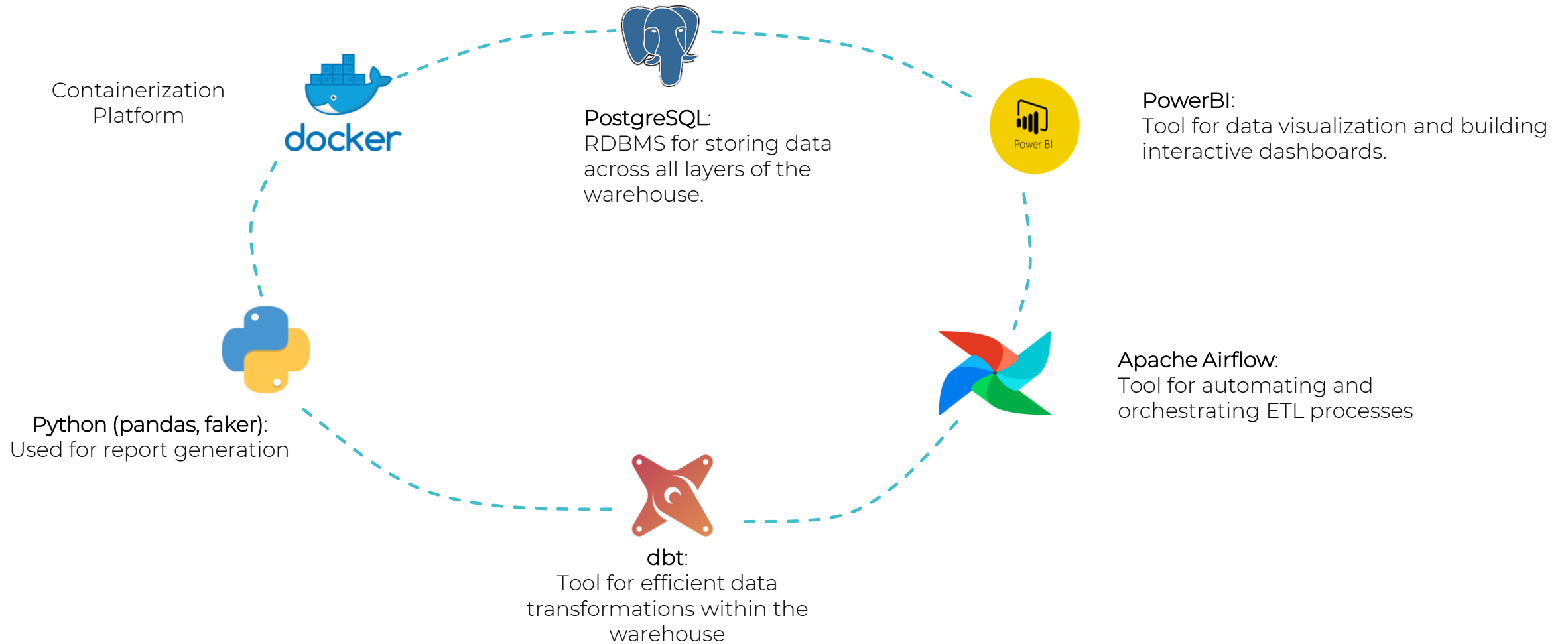
dbt: Alternative transformation
layer utilizing dbt capabilities

Metadata: Layer that stores
information about data loads and
ETL process results, helping to
track the status of updates and
loads.



Technology Stack

Slide 4



New data generation

Slide 5

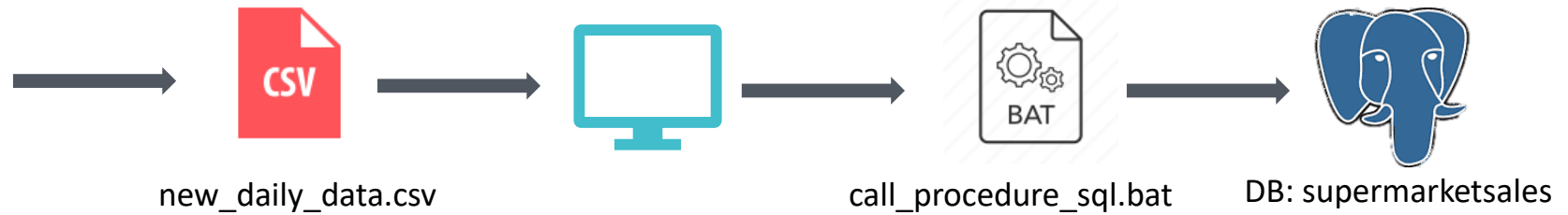
Objective:

To generate approximately 1200 new rows of data daily to simulate new arrivals and check data quality.

Process::

- 1.Generation in Docker — A Python script creates new_daily_data.csv and saves the data on the host
- 2.Automatic Loading into PostgreSQL — A .bat script calls a stored procedure to transfer data into the stage layer
- 3.The crontab is configured to run the script in the Docker container and invoke the .bat file

```
# m h dom mon dow  command
00 07 * * * /usr/bin/docker start daily_report
03 07 * * * /mnt/c/Windows/System32/cmd.exe /c "D:\Курсы и тренинги\Data Engineer Нетология\Дипломная работа\dataset\call_procedure_sql.bat"
```



Development of Data Warehouse Layers

Slide 6

Data Loading Stages (using Apache Airflow)

SELECT * FROM stage.kaggle_sales_ready

	invoice_id	branch	city	customer_type	gender	product_line	unit_price	quantity	tax_5	total	Date	Time	payment_type	cogs	gross_margin_percent
1	078-77-9200	A	Naypyitaw	Normal	Male	Fashion accessories	14,4	8	5,76	120,96	2019-12-21	08:20	Cash	115,2	4,7619
2	241-96-0209	A	Mandalay	Normal	Male	Health and beauty	79,02	5	19,75	414,85	2019-05-12	01:01	Cash	395,1	4,7619
3	973-11-2414	C	Mandalay	Normal	Female	Sports and travel	54,89	7	19,21	403,44	2019-10-11	10:82	Credit card	384,23	4,7619
4	934-37-2050	B	Yangon	Normal	Male	Electronic accessories	15,4	10	7,7	161,7	2019-05-14	02:03	Ewallet	154	4,7619
5	596-28-9626	B	Naypyitaw	Normal	Male	Health and beauty	62,65	9	28,19	592,04	2019-08-01	22:47	Ewallet	563,85	4,7619
6	820-22-6737	C	Yangon	Member	Male	Health and beauty	76,35	7	26,72	561,17	2019-05-26	14:21	Cash	534,45	4,7619
7	709-82-6363	C	Yangon	Normal	Male	Electronic accessories	6,76	4	-1,35	25,69	2019-05-13	11:33	Cash	27,04	4,7619
8	034-29-0384	C	Yangon	Member	Female	Sports and travel	48,07	2	4,81	100,95	2019-12-22	03:28	Ewallet	96,14	4,7619
9	672-29-4174	C	Naypyitaw	Member	Male	Home and lifestyle	12,36	8	4,94	103,82	2019-05-08	10:02	Credit card	98,88	4,7619
10	229-11-0367	A	Mandalay	Normal	Male	Electronic accessories	55,19	10	27,59	579,49	2019-12-26	04:07	Cash	551,9	4,7619
11	346-73-9386	C	Mandalay	Member	Male	Home and lifestyle	64,29	5	16,07	337,52	2019-08-01	15:33	Credit card	321,45	4,7619
12	012-19-5586	B	Naypyitaw	Member	Female	Sports and travel	80,56	3	12,08	253,76	2019-05-31	14:21	Ewallet	241,68	4,7619
13	363-01-8340	A	Naypyitaw	Member	Female	Health and beauty	39,62	4	7,92	166,4	2019-04-23	15:59	Credit card	158,48	4,7619
14	444-90-9633	A	Yangon	Member	Male	Sports and travel	27,68	2	2,77	58,13	2019-02-21	21:27	Ewallet	55,36	4,7619
15	028-47-6639	A	Mandalay	Normal	Male	Sports and travel	33,62	7	11,77	247,11	2019-06-04	23:05	Ewallet	235,34	4,7619

supermarketsales

- Схемы
 - dim
 - dq
 - fact
 - metadata
 - nds
 - stage
 - Таблицы
 - days_off 120K
 - kaggle_sales 176K
 - kaggle_sales_ready 2,5M

Stage

Raw data is loaded into the data warehouse

NDS

The data schema is transformed to comply with the 3NF

DDS

Final data loading into the multidimensional data warehouse, ready for analytics.

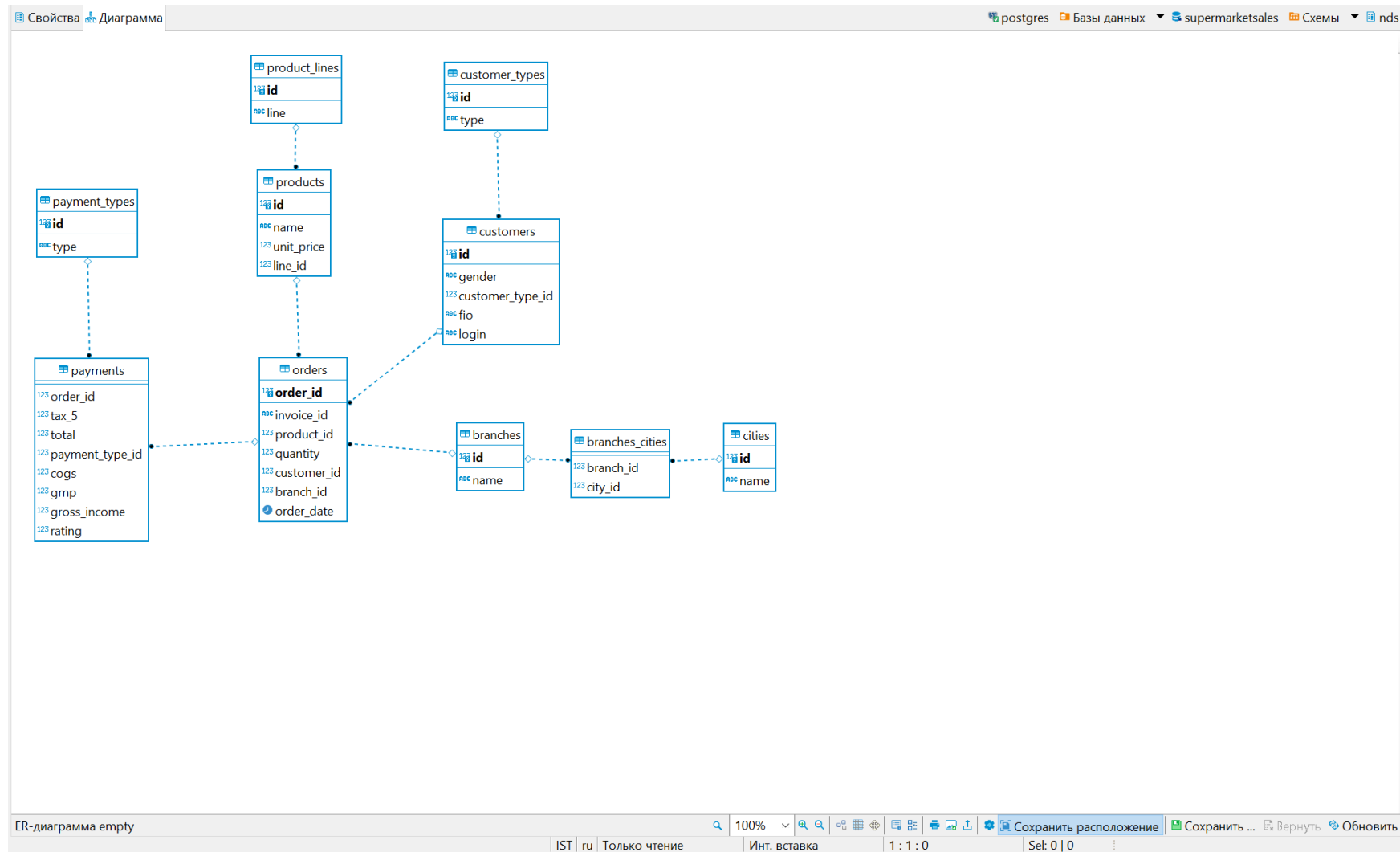
supermarketsales

- Схемы
 - dim
 - Таблицы
 - branches 40K
 - branches_cities 8K
 - cities 40K
 - customers 2,1M
 - dates 440K
 - functional_structure 56K
 - payment_types 40K
 - products 184K
 - Представления
 - Мат. представления
 - Индексы
 - Функции
 - Последовательности
 - Типы данных
 - Агрегатные функции
 - dq
 - fact
 - Таблицы
 - sales_items 1016K

Development of Data Warehouse Layers

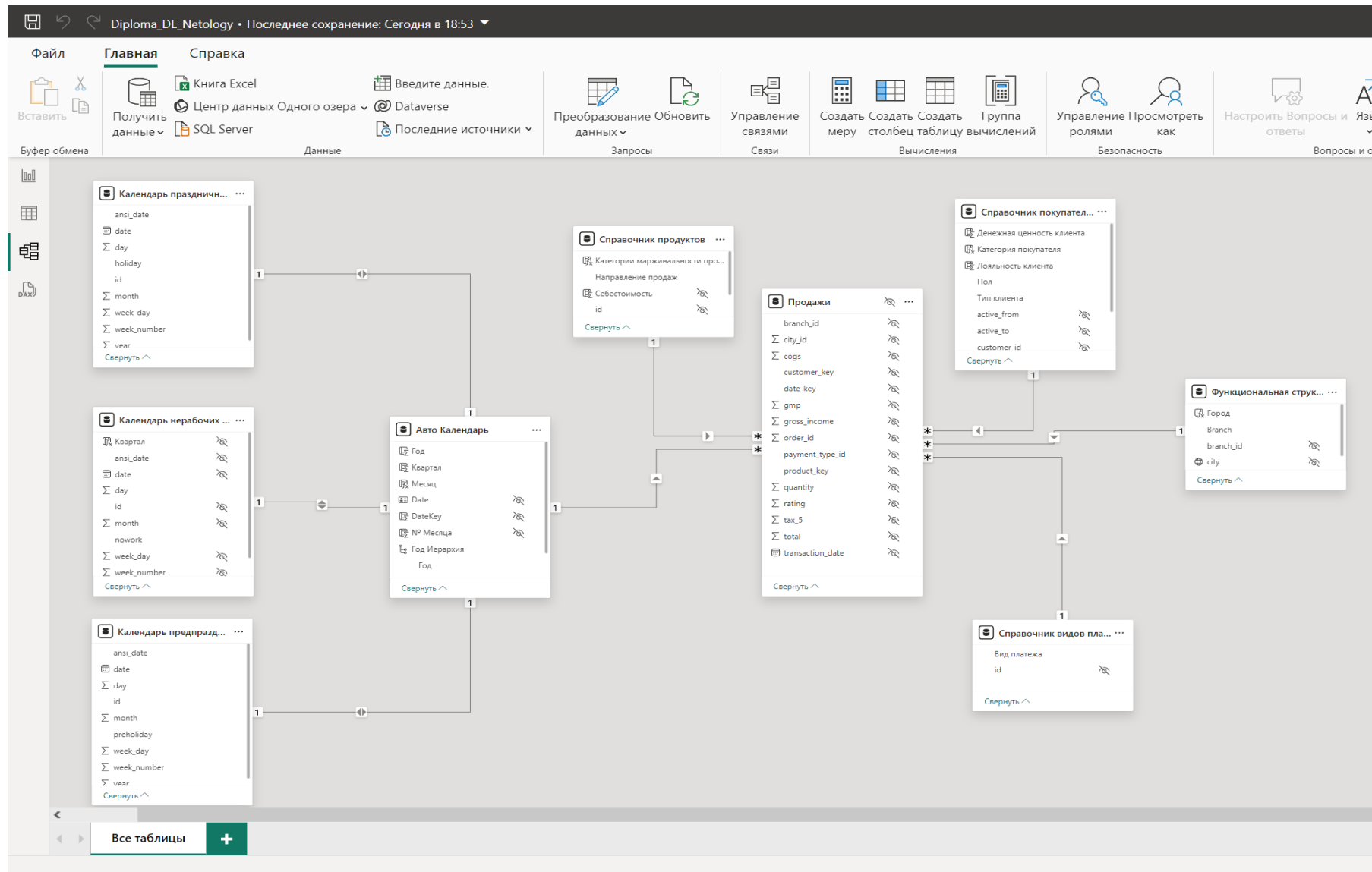
Slide 7

ERD for NDS schema



Data model in Power BI dashboards

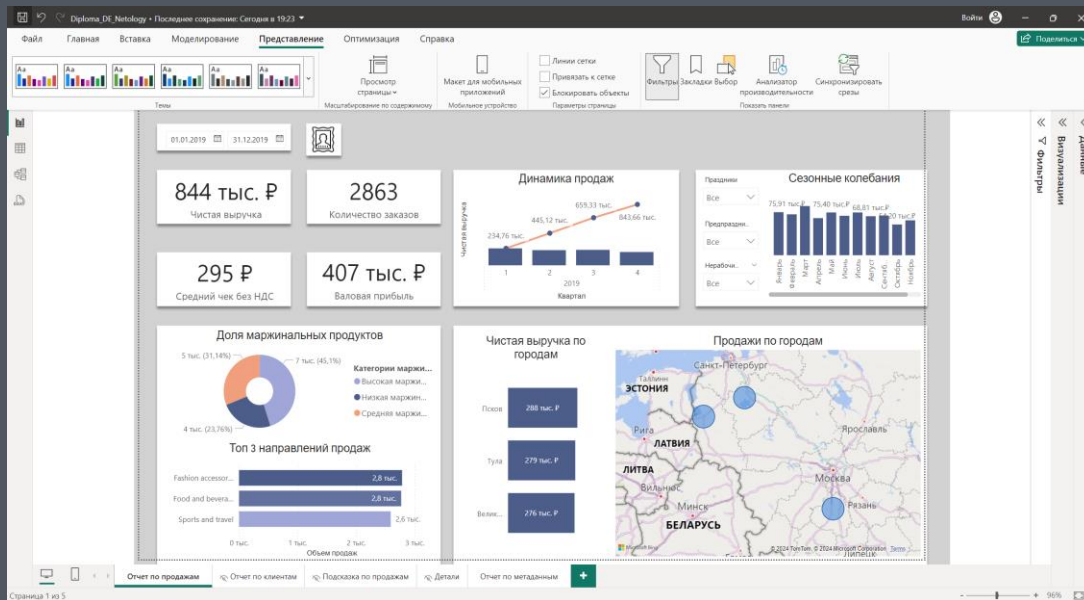
Slide 8



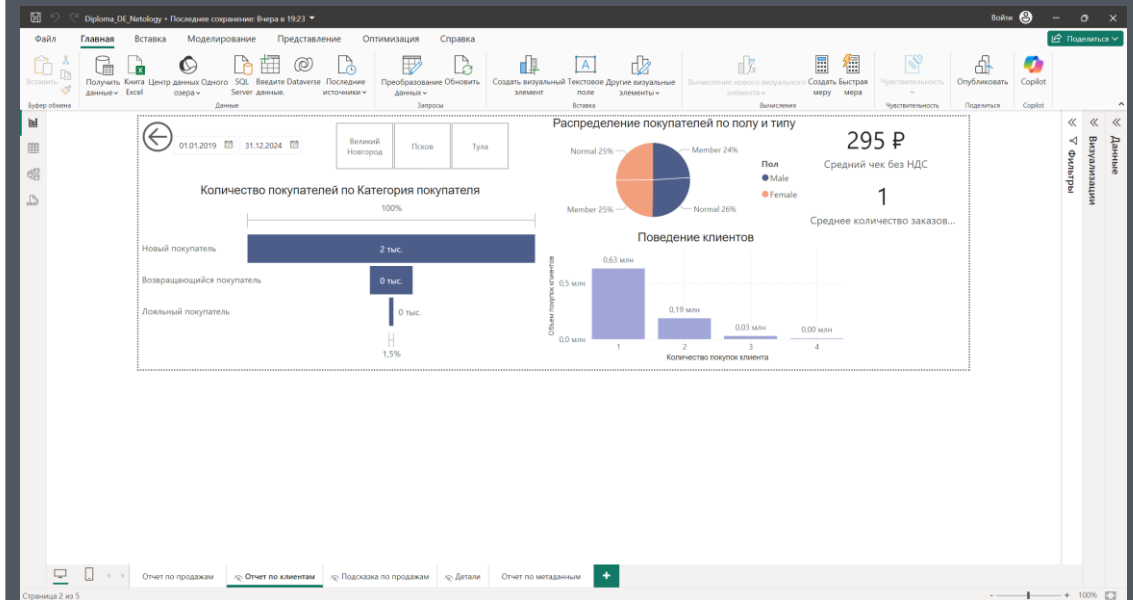
Examples of Dashboards and Metrics

Slide 9

Sales by Branch: Tracks sales volume by each branch, helping to identify the most profitable and underperforming stores.

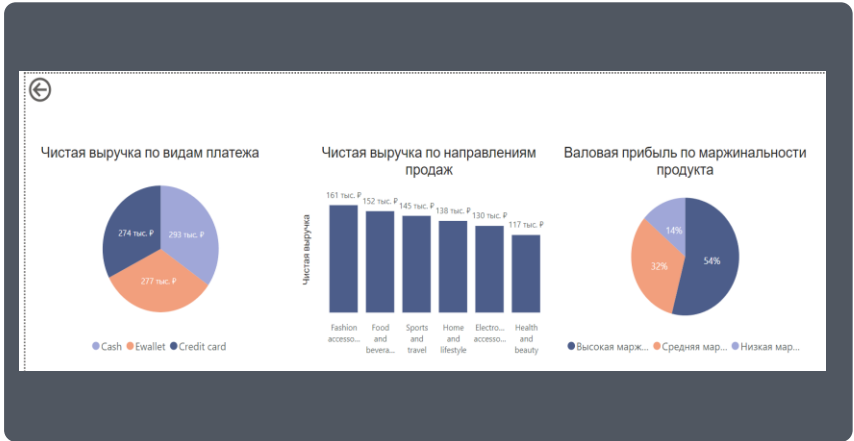
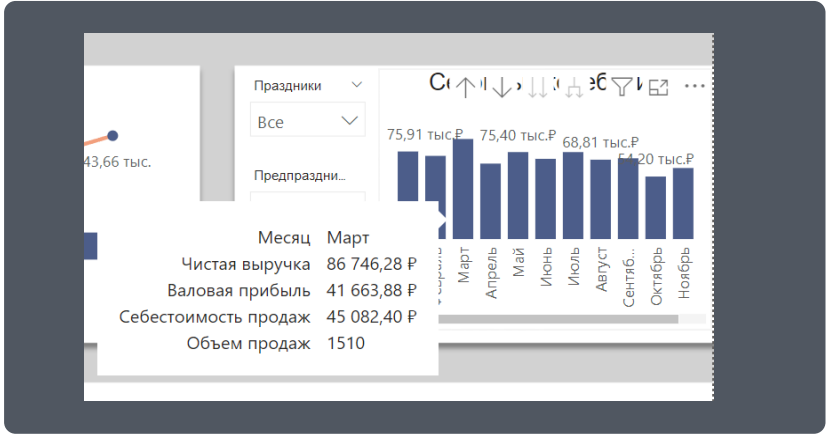
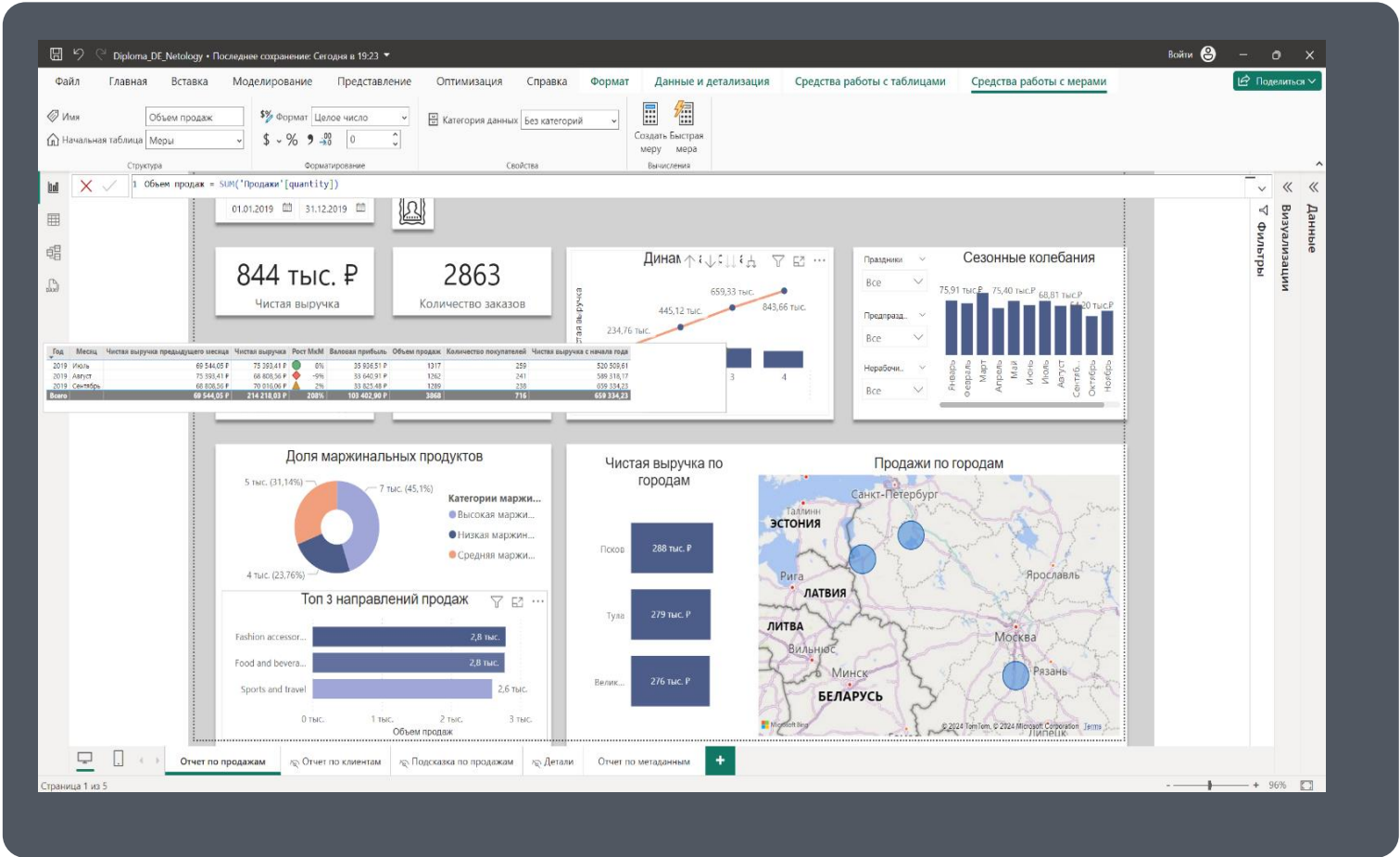


Customer Profile: Enables the analysis of customer behavior and forecasting based on customer preferences.



Examples of Dashboards and Metrics

Supplementary information on the Dashboard



ETL

Slide 11

Data Loading Stages (using Apache Airflow):

1

Data Quality Check (DQ)
in the Stage Layer

2

Loading Data into the
NDS Layer

3

Logging the Load to the
NDS Layer

4

Loading Data into the DDS
Layer

5

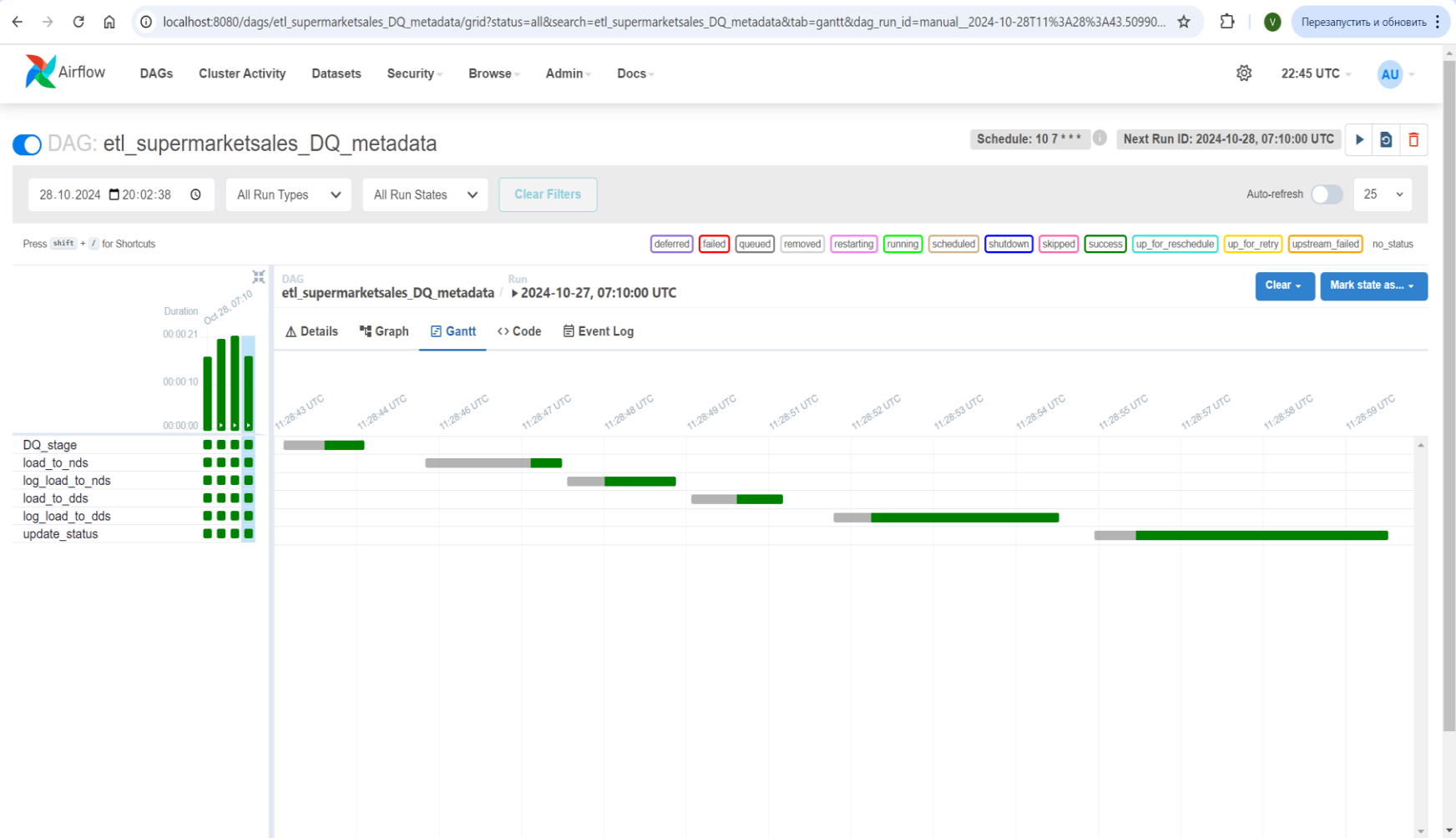
Logging the Load to the
DDS Layer

6

Updating Task Statuses in
the Metadata Table

ETL

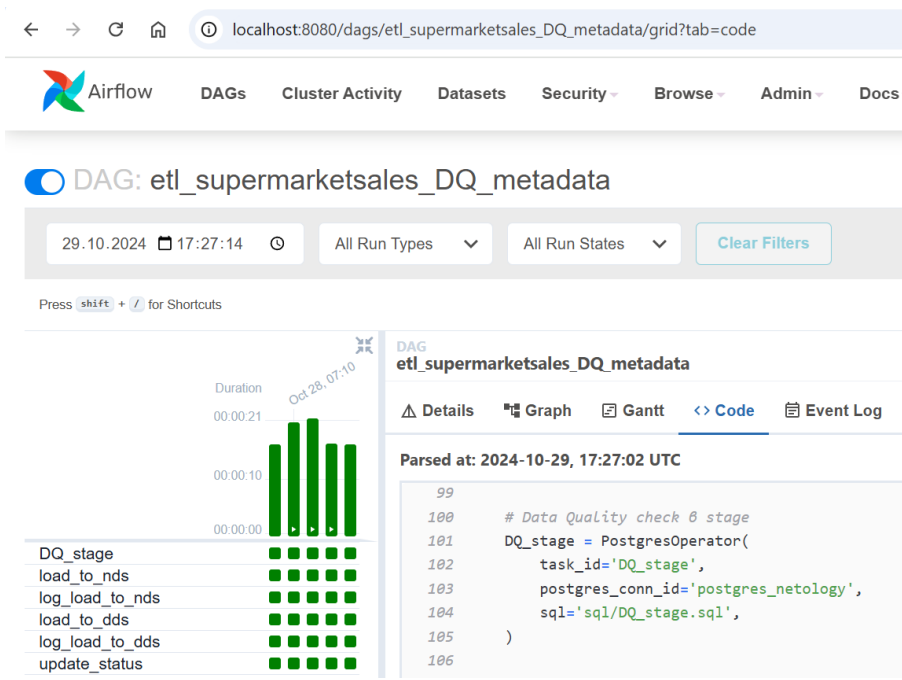
DAG etl_supermarketsales_DQ_metadata



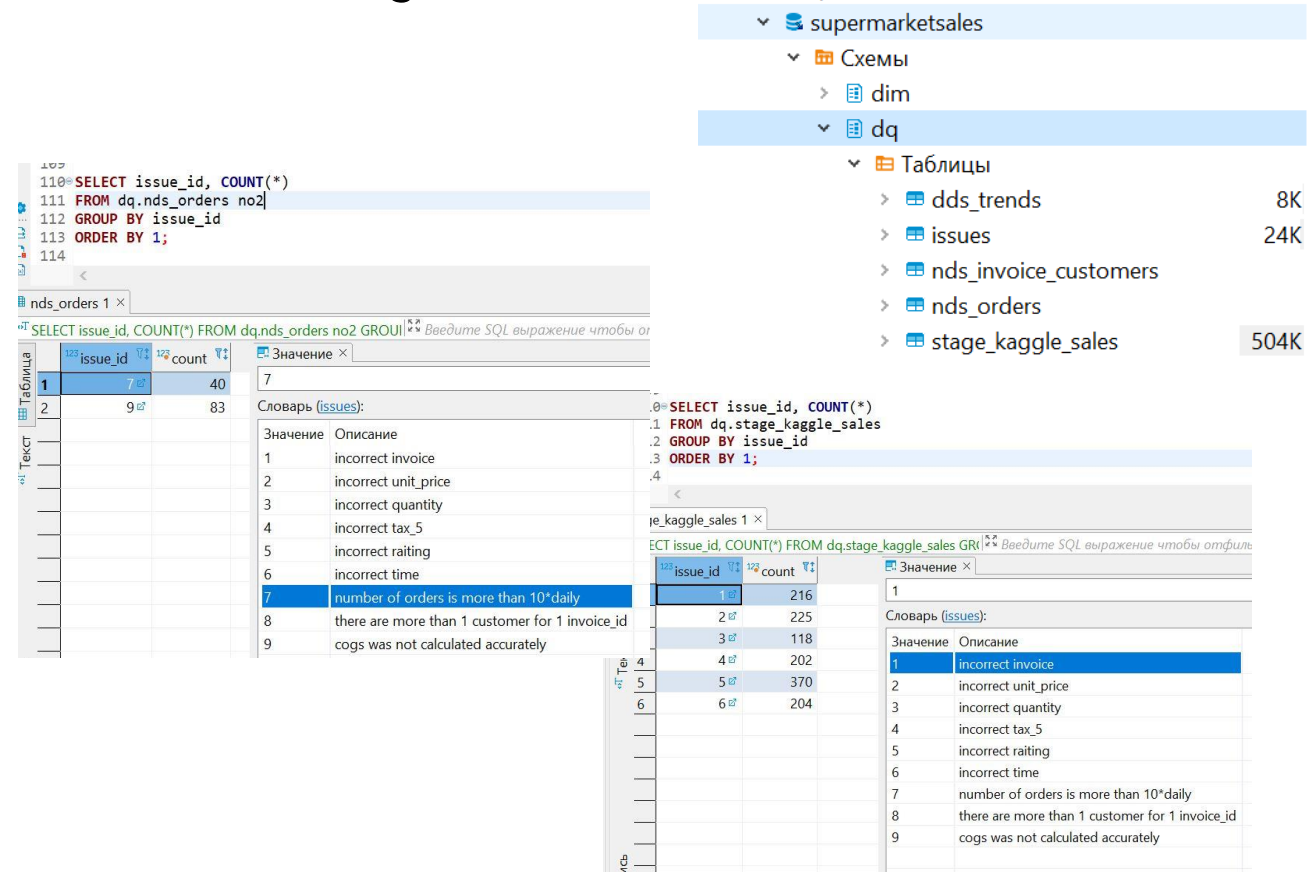
Data Quality Control

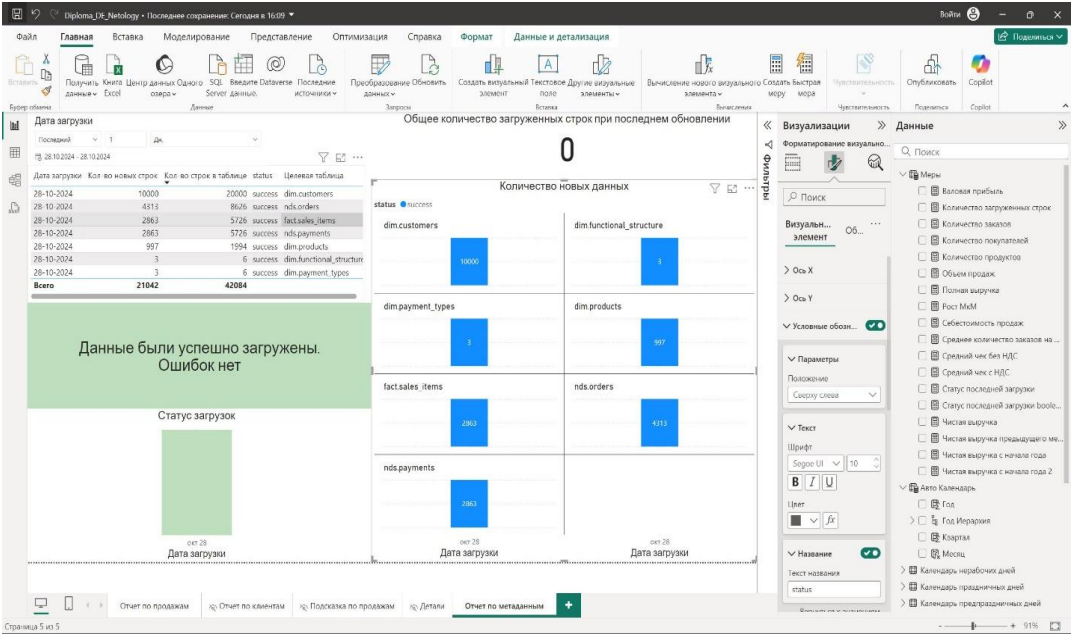
Approaches to Data Validation:

Data quality control is integrated into the ETL process



A separate framework has been established for handling invalid data





- Metadata for Tracking Loads:
- ✓ Records the status of loads (success/error), time, and data volume.
 - ✓ Ensures transparency of ETL processes and quick identification of issues.

Example of Load Status

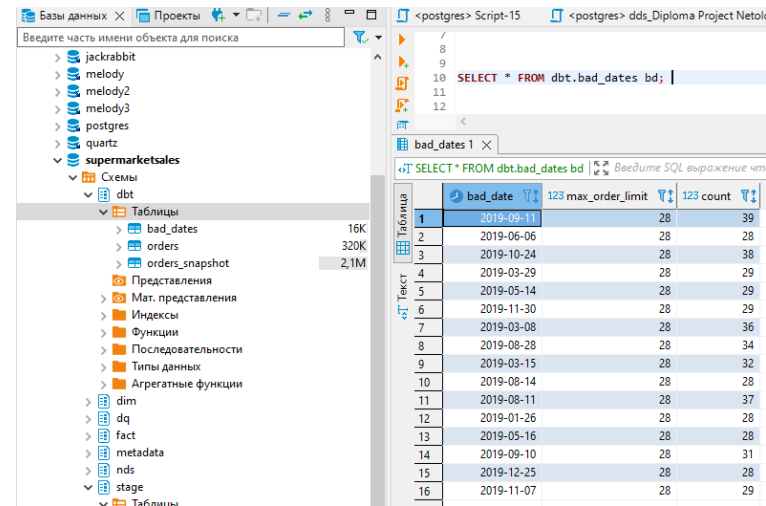
SELECT * FROM metadata.etl_status ORDER BY execution_date
Введите SQL выражение чтобы отфильтровать результаты

	123id	abc dag_id	task_id	execution_date	status	123rows_processed	target_table	123last_row_count	created_at
1	100	etl_supermarketsales_DQ_metadata	log_load_to_nds	2024-10-28 11:28:43.509	success	0	nds.payments	2863	2024-10-28 14:28:49.560
2	103	etl_supermarketsales_DQ_metadata	log_load_to_dd	2024-10-28 11:28:43.509	success	0	dim.payment_types	3	2024-10-28 14:28:53.857
3	101	etl_supermarketsales_DQ_metadata	log_load_to_dd	2024-10-28 11:28:43.509	success	0	fact.sales_items	2863	2024-10-28 14:28:53.268
4	99	etl_supermarketsales_DQ_metadata	log_load_to_nds	2024-10-28 11:28:43.509	success	0	nds.orders	4313	2024-10-28 14:28:49.360
5	102	etl_supermarketsales_DQ_metadata	log_load_to_dd	2024-10-28 11:28:43.509	success	0	dim.products	997	2024-10-28 14:28:53.453
6	104	etl_supermarketsales_DQ_metadata	log_load_to_dd	2024-10-28 11:28:43.509	success	0	dim.functional_structure	3	2024-10-28 14:28:54.065

dbt

– Tool for Convenient Transformation, Testing, and Documentation of Data in the Data Warehouse

- ✓ A single test model has been implemented, performing loading after checking outliers on the number of daily orders
- ✓ A Type 2 Slowly Changing Dimension (SCD) mechanism has been implemented for the dbt.orders table
- ✓ Documentation for the dbt project has been compiled



	bad_date	max_order_limit	count
1	2019-09-11	28	39
2	2019-06-06	28	28
3	2019-10-24	28	38
4	2019-03-29	28	29
5	2019-05-14	28	29
6	2019-11-30	28	29
7	2019-03-08	28	36
8	2019-08-28	28	34
9	2019-03-15	28	32
10	2019-08-14	28	28
11	2019-08-11	28	37
12	2019-01-26	28	28
13	2019-05-16	28	28
14	2019-09-10	28	31
15	2019-12-25	28	28
16	2019-11-07	28	29

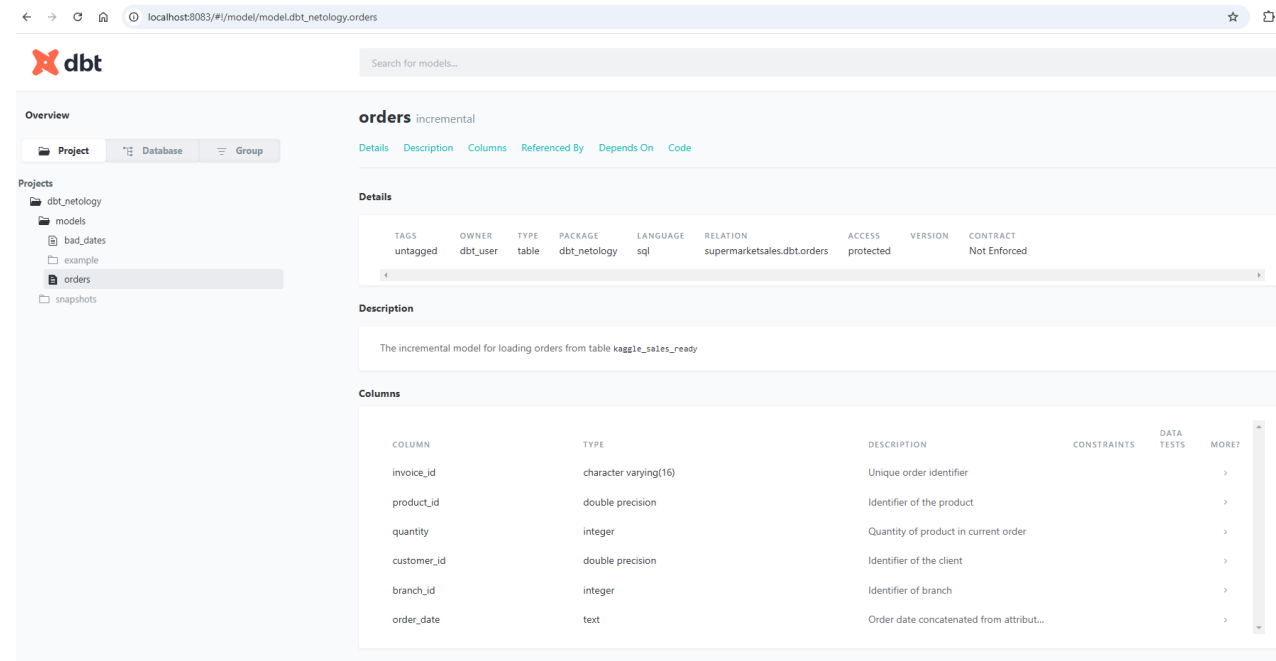
Slide 15

```
quantum@DESKTOP-VJG26RT: /mnt/d/Курсы и тренинги/Data Engineer Нетология/Дипломная работа/dbt/dbt_netology/models$ ls
bad_dates.sql  models.yml  orders.sql
quantum@DESKTOP-VJG26RT: /mnt/d/Курсы и тренинги/Data Engineer Нетология/Дипломная работа/dbt/dbt_netology/models$ cat orders.sql
{{ config(materialized='incremental', unique_key='invoice_id') }}

SELECT
    invoice_id,
    CEIL(random() * 997) AS product_id,
    quantity,
    CEIL(random() * 10000) AS customer_id,
    b.id AS branch_id,
    ("Date" || ' ' || "Time") AS order_date
FROM stage.kaggle_sales_ready d
JOIN nds.branches b ON b."name" = d.branch
WHERE NOT EXISTS (
    SELECT 1
    FROM dbt.orders o
    WHERE o.invoice_id = d.invoice_id
)
AND NOT EXISTS (
    SELECT 1
    FROM {{ ref('bad_dates') }} bd
    WHERE d."Date" = bd.bad_date
)

{% if is_incremental() %}
    -- Условие для инкрементальной загрузки
    AND d."Date" > (SELECT MAX(LEFT(order_date, 10)::DATE) FROM dbt.orders)
{% endif %}
ORDER BY 5, 6

(dbt-env) quantum@DESKTOP-VJG26RT: /mnt/d/Курсы и тренинги/Data Engineer Нетология/Дипломная работа/dbt/dbt_netology/models$
```



TAGS	OWNER	TYPE	PACKAGE	LANGUAGE	RELATION	ACCESS	VERSION	CONTRACT
untagged	dbt_user	table	dbt_netology	sql	supermarketsales.dbt.orders	protected		Not Enforced

COLUMN	TYPE	DESCRIPTION	CONSTRAINTS	DATA TESTS	MORE
invoice_id	character varying(16)	Unique order identifier			>
product_id	double precision	Identifier of the product			>
quantity	integer	Quantity of product in current order			>
customer_id	double precision	Identifier of the client			>
branch_id	integer	Identifier of branch			>
order_date	text	Order date concatenated from attribut...			>

Conclusions and Results

Slide 16

Goals Achieved:

- ✓ A multi-layered analytical data warehouse has been created
- ✓ Efficient ETL processes have been adjusted
- ✓ Dashboards have been developed to visualize key metrics

Thank you for your attention